

1. Basic Biology (Foundations)

Ex:

- Amino acids
 - Genes
 - Inheritance
 - Old couples → inherited traits → babies
 - Genetic disorders
 - How to test genetic issues
 - Applications of biology
 - Databases
 - How to access biological data
 - Protein structures
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2. Biology & Medicine Applications

Ex:

- How vaccines are developed
 - mRNA vaccines
 - Vaccine vs drug
 - Difference between:
 - Healthy people
 - Patients
 - Disease prevention
 - Future diseases
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3. Core Biological Facts

Ex:

- Number of nucleotides available
 - Number of chromosomes in humans
 - 23 pairs
 - 46 chromosomes
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4. Basic Biology with Computing

Ex:

- Biological data types:
 - DNA sequence data
 - Audio
 - Video
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5. Data Processing in Bioinformatics

Ex:

- Suggest solutions for biological data processing problems
 - Examples:
 - Converting ECG data to Excel
 - Handling biological datasets
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6. Sequence Analysis Concepts

Ex:

- Sequence assembly:
 - Breaking sequences into small pieces
 - Matching fragments
 - Proposing ideas/algorithms
 - Example rule: $A \rightarrow T$, $C \rightarrow G$
 - Reconstructing the original sequence from fragments
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7. Experimental Design & Analysis

Ex:

- Compare:
 - Healthy groups
 - Controlled groups
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8. Past papers

9. Biological Databases

- Ex:NCBI database
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10. Problem Solving & Computational Approaches

- Propose solutions for given problems using:
 - Machine Learning (ML)
 - Other computational methods
 - Algorithms & concepts:
 - Branch and Bound
 - Parallel computing
 - Graph coloring
 - Calculations
 - Data structures
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11. Programming & Methods

- Basic programming concepts
 - Calculations
 - Computational methods
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